Applied GIS

Land Sciences

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Applied GIS (A07605)

Short Title:	Applied GIS		
Faculty	Science and Computing		
Department	Land Sciences		
Cluster	Forestry		
Author	MPEDINI		
Credits	5	**Level:**	Introductory

Description of Module / Aims

In this module students will create thematic maps using a commercial GIS software package (ArcGIS). The module is very practical and will enable students to use online base maps, import georeferenced digital maps/ aerial photos, and create thematic maps relevant to forestry. Students will also be able to incorporate their own GPS survey data into the GIS maps. The module will equip the student with the skills required to create maps for the Forest Management Plan module in third year.

Programmes

GISS-0001	BSc in Forestry (WD_SFORS_D)	2 / 3 / M

Indicative Content

- Basic concepts of GIS
- ArcGIS and the file structure of ArcGIS projects (ArcCatalogue and ArcMap)
- Creating data frames and feature layers incl. using base maps
- Importing external data: digital maps, digital orthophotos, and GPS data
- Managing attribute table
- Creating and editing features
- Using the symbology tool
- Creating layouts and printing maps

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the key concepts of Geographical Information Systems (GIS).
2. Apply fundamental routines in ArcGIS, i.e. create a map document, set data frame properties and add layers, Draw and edit polygon, line, and point themes, cut polygons, manage attribute tables, use symbols and legends, create and print layouts to scale.
3. Apply back ground layers for heads-up digitising using ArcGIS base maps, OSI digital maps and orthophotos.
4. Complete maps that are targeted towards the end-user by proper use of symbology and map layout.
5. Display Irish GIS data downloaded from the Internet.
6. Use data from a GNSS receiver and convert the map projection to the Irish grid systems (IG and ITM).

Learning and Teaching Methods

- Key concepts will be described in a small number of lectures.
- The teaching of key skills will take place in an IT lab where all students will have access to the GIS software. During class time the lecturer will demonstrate the use of the GIS while students follow on their own PC. Notes and flash movies prepared by the lecturer will act as support.

- Students will prepare their assignment in their own time.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Assignment	30%	2,3,4
Assignment	30%	2,3,4
Assignment	40%	1,4,5,6

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out assessments. Failure to meet the objectives of assessments. Unable to carry out or submit independent study and research. Unable to present written work with attention to correct formatting, grammar and spelling.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on assessments. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped. Can present written work correctly formatted with minor grammar and spelling errors only.

50%-59%: Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out assessments. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.

60%-69%: Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design and current knowledge limitations when carrying out assessments. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.

70%-100%: Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on assessments. Show initiative, individual thought, and enthusiasm in self-study and independent learning.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Practical	48	
Independent Learning	87	

Essential Material

- "ArcGIS Desktop Resources." ESRI. http://desktop.arcgis.com/en/desktop/latest/main/map/mapping-and-visualization-in-arcgis-for-desktop.htm
- "Waterford IT Virtual Learning Environment." Applied GIS lecture notes. moodle.wit.ie

Supplementary Material

- Kennedy, M. _Introducing geographic information systems with ArcGIS: a workbook approach to learning GIS_. 3rd ed.. New Jersey, USA: Hoboken, 2013.
- Madej, E. _Cartographic design using ArcView GIS_. Albany, USA: Onword Press, 2001.

Requested Resources

- Room Type: Computer Lab